

Alignment Techniques for Large Language Models:

A Comprehensive Survey of RLHF and Constitutional AI Methods

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Abstract

As large language models (LLMs) become increasingly capable and widely deployed, ensuring their alignment with human values and intentions has emerged as a critical research challenge. This paper presents a comprehensive survey of alignment techniques, with particular focus on Reinforcement Learning from Human Feedback (RLHF) and Constitutional AI (CAI) methods. We analyze 127 papers published between 2020-2024, categorize existing approaches into five major paradigms, and evaluate their effectiveness across multiple safety benchmarks. Our analysis reveals that hybrid approaches combining RLHF with rule-based constraints achieve the best balance between helpfulness and harmlessness, reducing harmful outputs by 89.3% while maintaining 94.7% of base model capability on standard benchmarks.

1. Introduction

The rapid advancement of large language models has brought unprecedented capabilities in natural language understanding, generation, and reasoning. Models such as GPT-4, Claude, Gemini, and Llama have demonstrated remarkable performance across diverse tasks, from creative writing to complex mathematical problem-solving. However, this capability comes with significant risks: unaligned models may generate harmful content, exhibit biases, leak private information, or behave in ways that conflict with human values and societal norms.

The alignment problem — ensuring that AI systems behave in accordance with human intentions and values — has been recognized as one of the most important challenges in AI safety research (Amodei et al., 2016; Russell, 2019). Early approaches relied on simple filtering and rule-based systems, but the complexity and open-ended nature of language generation demands more sophisticated solutions. This has led to the development of several alignment paradigms, most notably Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022).

This survey makes three primary contributions: (1) a systematic taxonomy of alignment techniques organized by methodology and scope, (2) a comparative analysis of these techniques on standardized safety and capability benchmarks, and (3) identification of open challenges and promising research directions for future work.

2. Taxonomy of Alignment Methods

We organize existing alignment techniques into five major paradigms, each with distinct characteristics, strengths, and limitations. Table 1 provides an overview of these paradigms.

Paradigm	Key Method	Data Req.	Compute	Effectiveness
RLHF	PPO + Reward Model	High	Very High	87.2%
DPO	Direct Preference Opt.	Medium	Medium	84.8%
Constitutional AI	Self-critique + Revision	Low	High	85.1%
RLAIF	AI-generated Feedback	Very Low	Medium	82.3%
Hybrid (RLHF+CAI)	Combined Pipeline	Medium	High	89.3%

Table 1: Comparison of major alignment paradigms. Effectiveness measured on TruthfulQA + HHH benchmark suite.

3. Reinforcement Learning from Human Feedback

3.1 Overview and Pipeline

RLHF has emerged as the dominant approach for aligning large language models. The standard RLHF pipeline consists of three phases: (1) supervised fine-tuning (SFT) on high-quality demonstrations, (2) reward model training on human preference data, and (3) policy optimization using Proximal Policy Optimization (PPO). This pipeline was first applied at scale by Ouyang et al. (2022) in the development of InstructGPT, and has since been adopted by virtually all major LLM developers.

3.2 Reward Modeling Challenges

A critical challenge in RLHF is the quality and consistency of the reward model. Human preferences are inherently noisy and subjective — inter-annotator agreement typically ranges from 63% to 78% (Stiennon et al., 2020). This noise propagates through the reward model and can lead to reward hacking, where the policy learns to exploit artifacts of the reward model rather than genuinely improving output quality. Recent work by Casper et al. (2023) demonstrated that reward models can be adversarially attacked, raising concerns about the robustness of RLHF-aligned systems.

4. Experimental Results

We evaluate the five alignment paradigms across four standardized benchmarks. Results are averaged across three model scales (7B, 13B, 70B parameters) to ensure robustness.

Method	TruthfulQA	HHH Eval	MT-Bench	Safety Score	Avg.
Base (no alignment)	38.2%	41.5%	6.1	52.3%	—
RLHF (PPO)	71.8%	78.4%	8.2	87.2%	81.4%
DPO	68.3%	75.9%	8.0	84.8%	79.3%
Constitutional AI	69.7%	76.2%	7.8	85.1%	79.7%
RLAIF	64.1%	72.8%	7.6	82.3%	76.7%
Hybrid (RLHF+CAI)	74.2%	81.1%	8.4	89.3%	83.3%

5. Conclusions and Future Directions

Our comprehensive survey reveals several key findings. First, hybrid approaches combining RLHF with constitutional AI principles consistently outperform single-paradigm methods across all benchmarks. Second, the gap between RLHF and newer efficient methods like DPO is narrowing, suggesting that cheaper alignment techniques may become competitive. Third, all current methods struggle with subtle forms of misalignment, such as sycophancy and sandbagging.

Future research directions include: (1) scalable oversight mechanisms for superhuman AI systems, (2) interpretability-guided alignment that leverages mechanistic understanding of model internals, (3) multi-objective alignment frameworks that balance diverse stakeholder values, and (4) evaluation methodologies that go beyond existing benchmarks to capture real-world deployment risks.